

Higgs and Seek

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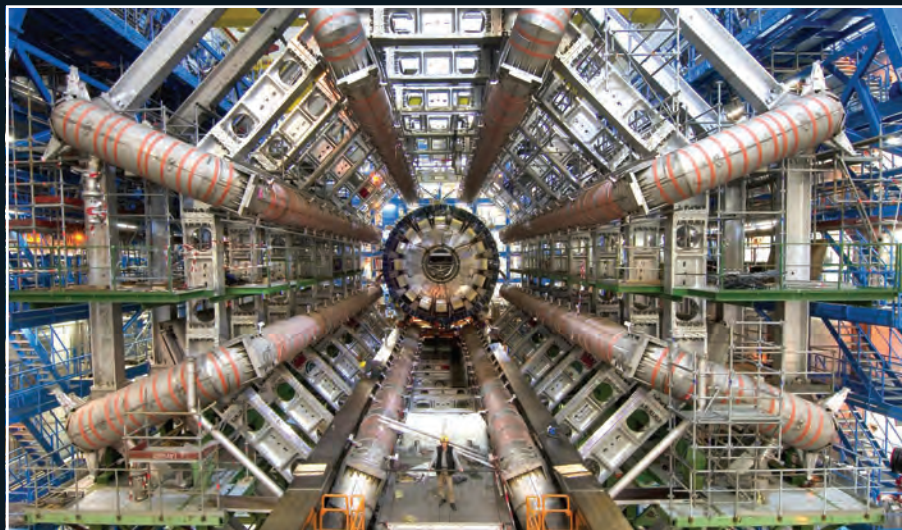
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Higgs Boson is a hypothetical fundamental particle theorized by Peter Higgs (that's how it got its name). What has made the entire physics community and the press go ga-ga over this particle is the fact that, if it were to exist, it would provide the mechanism which explains why everything has a mass - the sun, the tau neutrino and even you. It is considered to be a key particle for the Standard Model of Particle Physics.

In the strictest scientific terms, Higgs Bosons act as mediators for the Higgs Field. A Higgs field is something like an electromagnetic field. We're completely surrounded by Electromagnetic fields - radio signals, cell phones, satellite signals, radar all the time. It's an ocean of photons (a quantum of electromagnetic force) that completely blankets us. A Higgs field behaves in exactly the same way, it engulfs all the particles in an unseen ocean-like ether — permeating the entire universe. And Higgs bosons are the particles that constitute this field, just like H₂O molecules in water.

It's the interaction of particles with the Higgs field that causes drag and is manifested as mass. As a simple analogy consider a room filled with people, when a celebrity steps through the door to get to the other side of the room, people will cluster around her and make it difficult for her to move increasing the 'drag' and thus imparting more mass. When a relatively unimportant person like a member of the Digit editorial team steps into the room to get to the other side, his progress will be nearly uninhibited (nobody usually gives a hoot about what we do) and thus less mass. Hence it's the interaction of the elementary particles with the Higgs bosons that imparts mass to the particle.

Without mass, the universe would be a very different place. For example, if the electron had no mass, there would be no atoms and in a logical progression no sun, no earth and no people. It's the reason why elemen-



ATLAS is one of the six particle detectors setup at LHC that can find the Higgs Boson



A Top quark interacting with Higgs field

tary particles have mass and thus stars, planets and life exists. This is the reason why Higgs which imparts mass is called the 'God' particle. Though the physicists actually hate when people refer to Higgs boson as the 'God' particle. They consider it to be a hyperbole, all their efforts are directed to plugging in the holes in the standard particle model and thus understanding how the universe came into existence.

Scientists have setup the world's costliest experiment read LHC (Large Hadron Collider) at CERN in Switzerland, just to get a glimpse of this bashful particle. The LHC works by smashing together two beams of hadrons (in essence protons) travelling close to light speed. This generates a shower of particles that are only created at high energies. The Higgs will probably never be observed

directly, but scientists at the LHC hope that a Higgs will momentarily exist in this soup of particles. If it behaves as researchers think it will, it should decay further into yet more particles, leaving a trail that could prove its existence. Experiments carried out from 1989-2011 have ruled out the existence of Higgs outside the 116-130 GeV.

In December 2011 before closing for Christmas, LHC's two main experiments (ATLAS and CMS) both reported independently that their data hints at a possibility the Higgs may exist with a mass around 125 GeV/c². But they need more data to conclusively prove Higgs existence. A definitive statement on the existence or non-existence of the Higgs will require more data, and is not likely until later in 2012.

It seems we've painted a very rosy picture, but finding the Higgs boson is not guaranteed. All this discussion is based on a hypothesis. Finding the Higgs boson will open gates for many more important discoveries in physics like super-particles and dark matter, increasing our understanding of the universe. On the other hand, finding no Higgs boson will surely be a disappointment (not even considering the financial loss), but at the same time pave way for a new breed of theories that try and explain the Higgs mechanism. **d**